Nisarg Makwana

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Personal Profile

Advanced Mechanical Engineer (MSc) with expertise in Computational Fluid Dynamics (CFD), surrogate modelling, and numerical analysis. Experienced in simulating complex chemical diffusion and electromagnetic properties using high-level software. Highly skilled in collaborating with cross-functional academic and industry teams to translate simulation data into actionable design improvements. Proven track record in developing multiphase continuum models and optimising fluid dynamics for biomedical applications. Passionate about leveraging computational tools to solve complex thermal and fluid challenges.

Work Experience

Research Assistant

Queens University, Belfast | *June 2023 – Present*

- Used FEA software to simulate how different materials and shapes handle stress and fluid flow, reducing the need for expensive physical prototypes.
- Redesigned component geometries to improve performance while reducing material waste, ensuring designs are both cost-effective and sustainable.
- Developed multiphase continuum finite element models to simulate water diffusion and Nitric Oxide (NO) release in MOF-loaded polymer catheters.
- Designed and built experimental rigs to test diffusion and permeation properties of porous polymer-based materials.
- Collaborated with the University of St Andrews to combine computational modelling and experimental approaches for optimising MOF loading and catheter geometry.
- Applied surrogate modelling and optimisation algorithms to determine ideal MOF loading and tube dimensions for different applications.
- Gained experience in lab instrumentation, prototype design, finite element analysis, and polymer systems for biomedical applications.

Researcher Intern

University of Leeds | *October 2022 – Present*

- Created complex 3D CAD models and performed flow analysis to check for potential design flaws or mechanical interference.
- Analysed mechanical parts to identify weak points, recommending changes to the geometry to improve product life and safety.
- Prepared clear design reports and 3D visualisations to help management and production teams understand technical requirements.
- Performed simulations, modelling, and numerical analysis of the computational model to evaluate the potential performance of proposed geometry changes.
- Prepared weekly reports and presentations detailing new findings, presented to company management and research supervisors using tables, graphs, and fact sheets.
- Optimised existing designs of the flow cell and improved product performance in terms of fluid mixing by 145%.

Engineering Intern

Indian Institute of Technology, Guwahati | *June 2024 – August 2025*

- Developed and verified analytical models for a 12/8 Bearing-less Switched Reluctance Motor (BSRM) using the Maxwell Stress Tensor method to calculate torque and controllable radial forces.
- Performed 2D Finite Element Analysis (FEA) using Ansoft Maxwell to validate mathematical models under static and transient conditions.
- Collaborated on the assembly of a prototype system incorporating displacement sensors, absolute encoders, and real-time controllers in LabVIEW for bearingless operation testing.
- Integrated FEA models with external circuit editors to analyse the performance of specialised asymmetric bridge converters during transient motor operation.
- Helped hypothesise and develop a test rig following IEC and ISO standards, featuring a 0.75 kW power output.

Leadership & Activities

Course Assistant / Demonstrator

Queens University, Belfast | *September 2024 – December 2025*

- Provided in-class technical and mathematical support as a Demonstrator for the Mathematics and Computing module.
- Conducted hands-on demonstrations of computing techniques including programming tools and mathematical software.

Visiting Researcher

University of Leeds | *April 2024 – Present*

- Executed Multiphysics modelling and simulation of a microfluidics project using COMSOL Multiphysics for analysis and optimisation.
- Facilitated production of research materials by documenting simulation methodologies and results.
- Validated and refined existing computational models, ensuring accuracy and reliability of simulations.
- Strategised with management, supervisors, and product designers to create 3D models for flow analysis and concept substantiation.

Education

M.Sc. Advanced Mechanical Engineering

University of Leeds, United Kingdom | *2021 – 2022*

- **Relevant Coursework:** Advanced Finite Element Analysis, Engineering Computational Methods, Team Design Project, Tribology and Surface Engineering, Computational Fluid Dynamics.
- **Design Project:** Flow analysis of a micromixer in liquid flow cell for Transmission Electron Microscope (TEM) based study.

B.Tech. Mechanical Engineering

Sardar Vallabhbhai National Institute of Technology, India | *2017 – 2021*

- **Relevant Coursework:** Plastics and Ceramics, Basic Mechanical Systems, Plant and Layout and Material Handling, Advanced Refrigeration and Air Conditioning, Theory and Design of Cryogenic Systems.
- **Design Project:** Identification of two-phase flow using Acoustic sensor.

Skills

Category	Details
Projects	Street lighting system (supply, delivery, installation, commission & maintenance); Flow analysis of a micromixer in liquid flow cell for TEM-based study
Programming	C, C++, Python, MATLAB, Pandas, LaTeX, Fortran, Gnu Octave, Scilab
Operating Systems	GNU/Linux, Windows (95, 98, XP, 7, 10)
Software Tools	SolidWorks, AutoCAD 2D, COMSOL Multiphysics, Ansys Workbench, FEAST, Altair Hyperworks, SimScale, Autodesk Inventor, Fusion 360, PTC Creo, Siemens NX, Catia V5, OpenFOAM, Simulink, ABAQUS CAE, Microsoft Office
Hands-on Experience	3D Printing, SEM Imaging, DSC, TGA, FTIR, CAD/CAM, CNC, Material Testing, Instron (tensile & compression testing)
Languages	English, Hindi, Gujarati

Certificates

- Data Analysis in Python — Jovian.ml
- Machine Learning — Stanford University
- FEAST Software Workshop (Finite Element Analysis of Structures) — ISRO
- SolidWorks & AutoCAD 2D Certified — RCAD Institute
- Aerial Robotics — University of Pennsylvania

No visa sponsorship required now or in future (currently on Global Talent Visa).